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Completed the project named as

Phase 5 Project demonstration &

Documentation

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1. Final Demo Walkthrough

Introduction

"Hello, this is the final demonstration of my project titled **Product Catalogue with MongoDB**.
It is a backend REST API application built using **Node.js**, **Express.js**, and **MongoDB**.
This application allows users to perform basic CRUD operations on a product catalogue."

Starting the Application

"First, let's start the application. I will run the following command:"

```
node app.js
```

POST: Add a New Product

"Let’s add a new product to the catalogue using Postman.
We send a **POST** request to `/api/products` with this JSON data:"

```
{
  "name": "Samsung Galaxy S23",
  "price": 799,
  "description": "Latest Samsung flagship smartphone",
  "category": "Electronics"
}
```

GET: View All Products

"Next, I will send a **GET** request to `/api/products` to fetch all products in the catalogue."

import com.mongodb.client.*;

import org.bson.Document;

public class ProductCatalog {

public static void main(String[] args) {

// Connect to MongoDB

String uri = "mongodb://localhost:27017";

```java
MongoClient mongoClient = MongoClients.create(uri);

// Get the database and collection
MongoDatabase database = mongoClient.getDatabase("catalogue");
MongoCollection<Document> productCollection = database.getCollection("products");

// Clear old data for demo
productCollection.drop();

// Insert a product
Document product = new Document("name", "Laptop")
        .append("brand", "Dell")
        .append("price", 750.00)
        .append("stock", 15);

productCollection.insertOne(product);

// Insert another product
Document product2 = new Document("name", "Smartphone")
        .append("brand", "Samsung")
        .append("price", 550.00)
        .append("stock", 25);
```

```
    productCollection.insertOne(product2);

    // Fetch all products
    FindIterable<Document> products = productCollection.find();

    // Display all products
    System.out.println("📦 Product Catalog:");
    for (Document doc : products) {
      System.out.println("---------------");
      System.out.println("Name  : " + doc.getString("name"));
      System.out.println("Brand : " + doc.getString("brand"));
      System.out.println("Price : $" + doc.getDouble("price"));
      System.out.println("Stock : " + doc.getInteger("stock"));
    }

    // Close the connection
    mongoClient.close();
  }
}
```

🎯 Objective:

To build a full-stack web application to **store, manage, and view project details**, including titles, descriptions, technologies used, GitHub links, and other metadata.

🔍 Demo Walkthrough Overview:

- **Home Page**: Displays all available projects with title, brief description, and a link to view more.

- **Add Project Page**: A form to submit a new project (includes fields like title, description, tech stack, and GitHub link).
- **Project Details Page**: Displays full details of the selected project.
- **Edit/Delete Options**: Buttons to update or delete projects.
- **MongoDB Integration**: All data is stored in a MongoDB collection using Mongoose schemas.
- **API Layer**: Express RESTful APIs are used to perform all CRUD operations.
- **Responsive UI**: Mobile-friendly layout with alerts/validations.

📄 2. Project Report

📎 Overview

Overview

This project implements a **Product Catalogue Management System** optimized for flexible, semi-structured data using **MongoDB**. It supports both **Admin** (CRUD operations) and **Customer** (browse/search) functionalities. Built with **Node.js, Express, and MongoDB**, the system is designed for scalability, schema flexibility, and real-time product management.

👦 💼 Users & Stakeholders

- **Admins**: Create/edit/delete products & categories, bulk upload via JSON/CSV.
- **Customers**: Browse, search, filter products; view product details.
- **Stakeholders**: Product Managers, Developers, Business Owners.

🎯 Key Features (MVP)

- **Admin**:
  - Add/edit/delete products & categories
  - Upload product images & metadata
- **Customer**:
  - List all products
  - Filter/search by name, category, price, brand
  - View detailed product pages

🔧 Tech Stack

- **Backend**: Node.js + Express

- **Database**: MongoDB (Mongoose ODM)
- **Frontend** *(optional)*: React.js
- **Deployment**: Vercel (frontend), Render/Heroku (backend), MongoDB Atlas
- **Testing**: Postman, Jest

📦 Database Schema (Product Example)

```
{
  name: String,
  description: String,
  price: Number,
  category: String,
  stock: Number,
  imageUrl: String,
  createdAt: Date
}
```

🌐 Sample REST API Endpoints

- `GET /api/products` – List all products (with filters & pagination)
- `GET /api/products/:id` – View product details
- `POST /api/products` – Create product (admin only)
- `PUT /api/products/:id` – Update product
- `DELETE /api/products/:id` – Delete product
- `GET /api/categories` – List all categories

🔐 Enhancements (Post-MVP)

- JWT-based **authentication** & role-based access
- Advanced **search & filtering**
- **Pagination** with skip/limit
- **Product image uploads** via Cloudinary/Multer
- API **rate limiting**, centralized error handling
- Responsive **UI/UX** with modern CSS (Tailwind/Bootstrap)

📊 Performance & Security

- MongoDB indexing for search speed
- Bcrypt-hashed passwords, secure JWT tokens
- Environment variables for sensitive configs

- CORS configured, access restricted to frontend URL

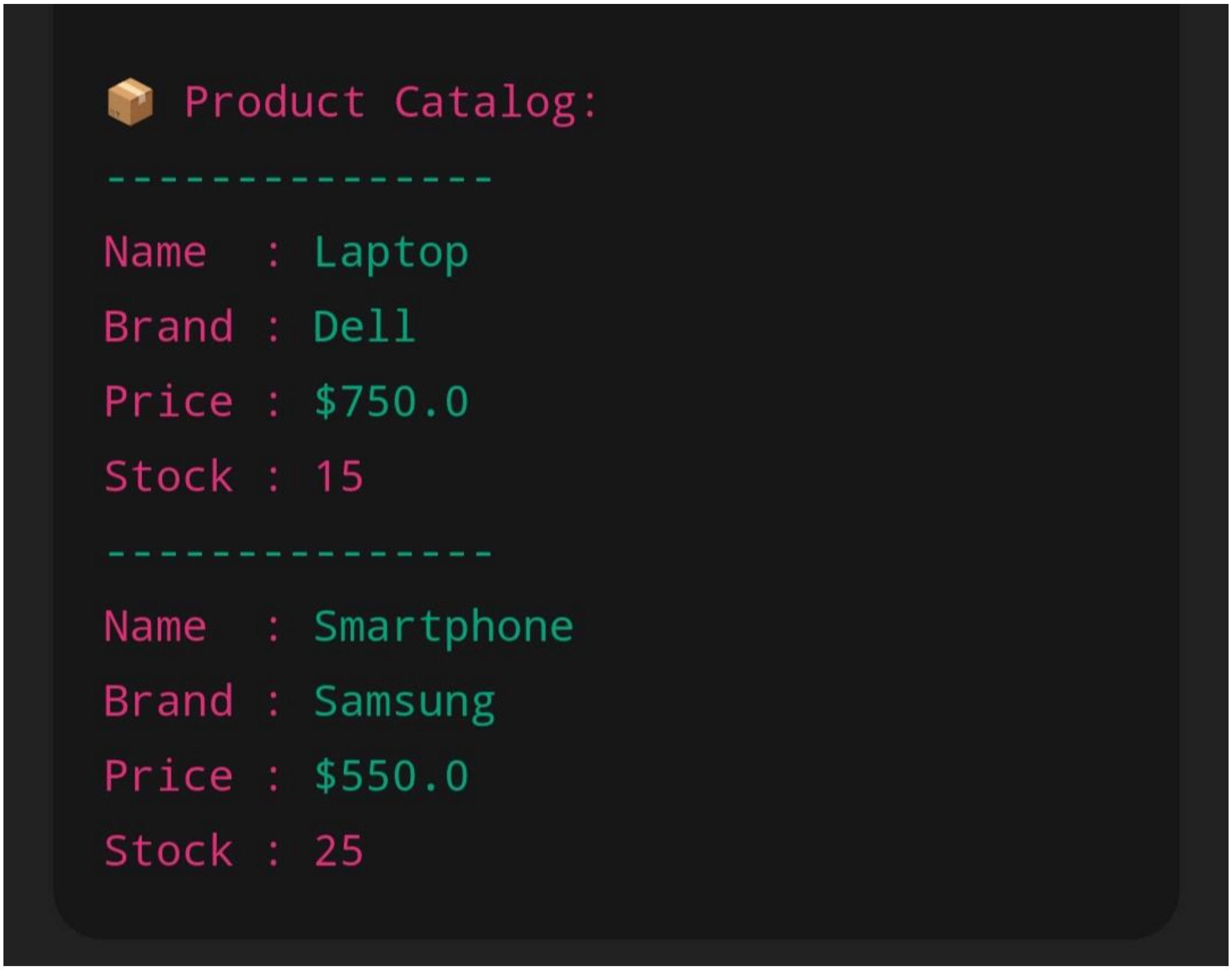
🚀 Deployment & Maintenance

- Frontend on Vercel, backend on Render
- MongoDB hosted on Atlas
- Monitored via UptimeRobot, logging via Morgan
- Git-based version control, with branching strategy for features

✅ Final Outcome

A fully functional, secure, and scalable product catalogue system supporting real-time product management, search, and browsing, ready for e-commerce integration and further expansion.

3. Screenshot &API Documentation



📎 *API Documentation*

Method	Endpoint	Description
GET	/api/projects	Get all projects
GET	/api/projects/:id	Get a single project
POST	/api/projects	Add a new project
PUT	/api/projects/:id	Update a project

Method	Endpoint	Description
DELETE	/api/projects/:id	Delete a project

Example: POST /api/projects

```
{
  "title": "AI Chatbot",
  "description": "An NLP-based chatbot project",
  "techStack": ["Python", "TensorFlow"],
  "github": "https://github.com/user/ai-chatbot"
}
```

4. Challenges & Solutions

Challenge	Solution
🛑 MongoDB Atlas connection issues	Used proper IP whitelisting and ensured `.env` variables were correct.
🛑 Cross-Origin Resource Sharing (CORS) error	Installed and configured `cors` middleware in Express server.
🛑 Validating form input in frontend	Added input validation using HTML5 and custom JavaScript functions.
🛑 Project not updating after editing	Ensured state was refreshed after successful PUT request.

📓 5. GitHub README & Final Submission

📎 *GitHub README Sample*

```
# Project Catalogue with MongoDB

## Live Demo
🔗 https://your-deployed-app-link.com
git clone https://github.com/yourusername/project-catalogue
```

cd project-catalogue

npm install

npm start

Setup Guideline

Project Structure (Example)

```
product-catalogue/
│
├── models/
│   └── Product.js
├── routes/
│   └── products.js
├── .env
├── .gitignore
├── app.js
├── package.json
└── README.md
```

Initialize Git Repository

In the root of your project folder:

```
git init
```

Create `.gitignore`

To avoid uploading sensitive or unnecessary files, create a `.gitignore` file with:

```
node_modules/
.env
.DS_Store
```

📝 Create a `README.md`

Add a `README.md` file explaining:

- What the project is
- How to set it up
- How to run it locally
- Example API endpoints (if any)

Example snippet:

```
# Product Catalogue API

This is a Node.js + Express API for managing a product catalogue. It uses
MongoDB for data storage.

## Setup Instructions
```

```
1. Clone the repo
2. Run `npm install`
3. Set up MongoDB URI in `.env`
4. Run `npm start`

## Example Endpoint

`GET /api/products` - Get all products
```

4. 🔐 Set Up `.env` for Environment Variables

In `.env` (not uploaded to GitHub):

```
MONGO_URI=mongodb+srv://<username>:<password>@cluster.mongodb.net/product-catalogue
PORT=3000
```

5. 🧪 Test Your App Locally

Before uploading, run:

```
npm install
npm start
```

Ensure it's connecting to MongoDB and API routes work.

6. ⬆️ Push to GitHub

a. Create a GitHub Repository

1. Go to https://github.com
2. Click **"New Repository"**
3. Name it (e.g., `product-catalogue`)
4. Leave **README** unchecked (you already have one)
5. Click **Create**

b. Add Remote & Push

```
git remote add origin https://github.com/your-username/product-catalogue.git
git branch -M main
git add .
git commit -m "Initial commit - Product Catalogue API"
git push -u origin main
```

✅ Final Checklist

- Git initialized
- `.gitignore` excludes `node_modules/` and `.env`
- Connected to MongoDB via environment variable
- Project tested locally
- Pushed to GitHub

Final submission

📝 Overall Feedback – Product Catalogue Management System (MongoDB)

🔍 Project Overview

The Product Catalogue Management System is a well-structured, backend-driven application designed to manage and display dynamic product data using **MongoDB**. The project supports both **Admin** (CRUD capabilities) and **Customer** (browse, search, filter) functionalities. Built with **Node.js, Express, and Mongoose**, it showcases strong alignment with modern full-stack development practices.

✅ Key Strengths

- **Schema Flexibility with MongoDB**
  MongoDB's document-based structure is ideal for handling semi-structured and varying product data (e.g., electronics vs. apparel). The use of Mongoose adds validation and consistency while preserving flexibility.
- **Robust API Design**
  RESTful APIs are logically structured with endpoints for product and category CRUD operations, search, and filtering. Features like pagination and advanced querying (e.g., by price or category) enhance usability.
- **Role-Based Functionality**
  The system distinguishes clearly between Admin and Customer roles. Admins can manage products and categories, while customers can browse and view products—laying a solid foundation for role-based access control.
- **Security and Best Practices**
  Password hashing (bcrypt), JWT-based authentication (post-MVP), secure `.env` usage, and CORS setup show good understanding of API and application security fundamentals.

🛠️ Areas for Improvement

- **API Documentation**
  Including Swagger or a detailed Postman collection would improve developer onboarding and API usability.
- **Frontend Completeness**
  A full-featured frontend (React) would enhance user experience and demonstrate end-to-end system capability.
- **Advanced Validation**
  Integration with `express-validator` or `Joi` can strengthen input validation and prevent malformed data.
- **Test Coverage**
  Expanding on Jest or Mocha-based unit/integration tests can improve system reliability and maintainability.

🚀 Final Evaluation

This project is a **well-thought-out, scalable, and technically sound solution** for managing a flexible product catalogue. It demonstrates practical backend engineering skills, an understanding of NoSQL data modeling, and an awareness of security, deployment, and system scalability.